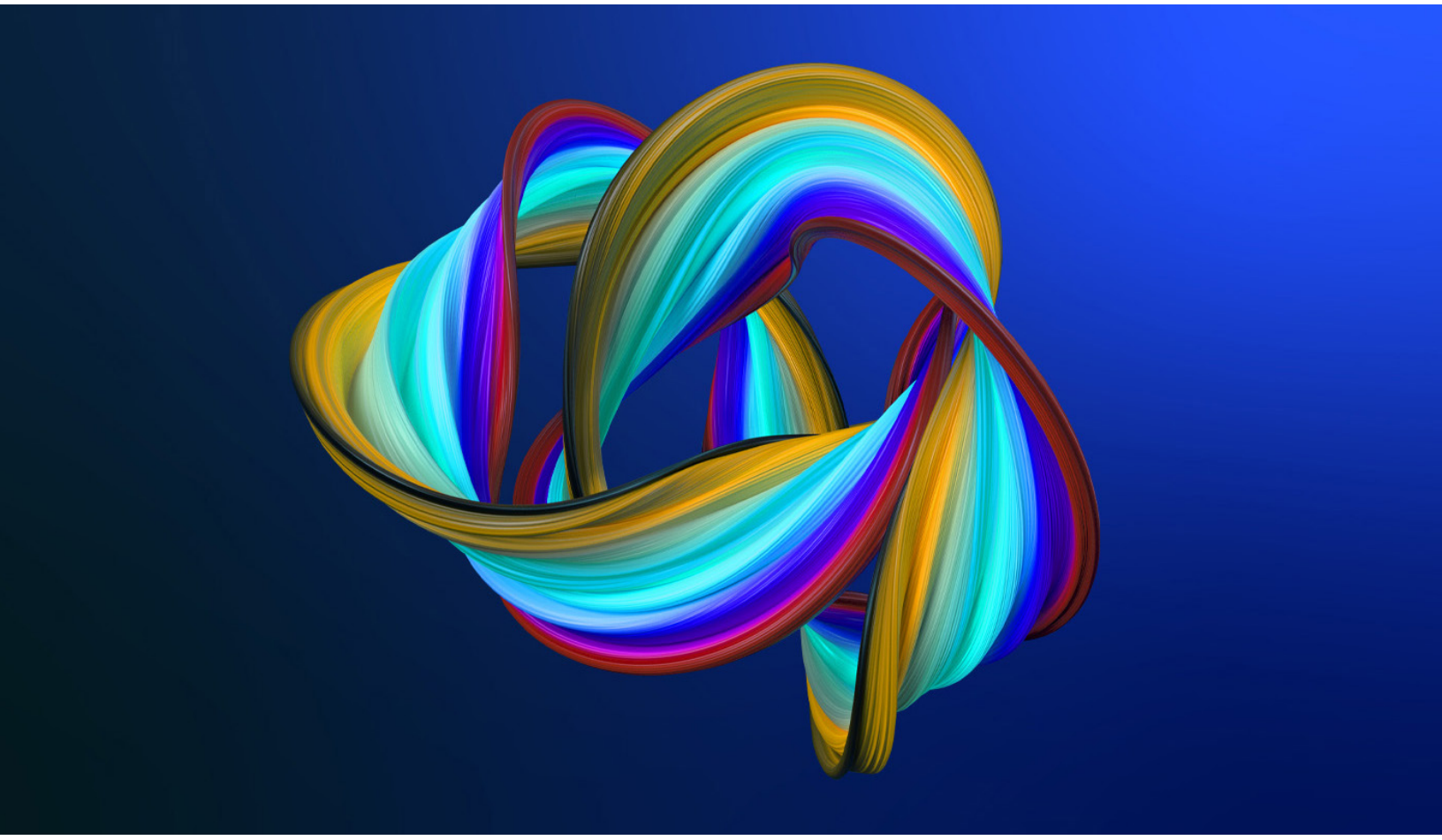
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The great trade rearrangement

A newly defined “rearrangement ratio” gauges the potential to rearrange US imports from China and how this could ripple across the globe.

by Olivia White, Jeffrey Condon, Chris Bradley, Michael Birshan, Sven Smit, and Camillo Lamanna



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At a glance

— **Amid pressure on US–China trade, firms may look to *rearrange* sourcing to alternative suppliers**. If they cannot, firms might instead *reduce* purchases, *replace* imported products with something similar, or *ramp up* domestic production. These alternatives require a combination of sacrifice, resources, know-how, and time.

— **We introduce a "rearrangement ratio" to quantify how hard the change might be.** Thirty-five percent of US imports from China have a ratio less than 0.1, signifying a global available export market ten times larger than current US imports from China—think T-shirts or logic chips. For higher ratios, rearrangement becomes harder, and for the 5 percent of trade with a ratio greater than 1.0—for example, rare earth magnets—US imports from China exceed available global exports.

— **Consumer goods are harder to rearrange than business inputs.** Sixty-one percent of business input imports have a rearrangement ratio less than 0.1, versus 16 percent of consumer goods. Major products like laptops, smartphones, and toys are harder to rearrange.

— **Europe emerges as the fulcrum of trade rearrangement.** Across nine varied simulations, European imports from China and exports to the United States both go up by nearly $200 billion. As intra-European trade shifts to the United States, it leaves holes filled by increased Chinese exports—assuming Europe does not choose to alter its own trade policies. Others will be affected, too: exports to the United States from as many as 70 countries may increase by more than 10 percent.

— **Prepare for resilience in a reordering world.** Strategies will need to handle continued uncertainty and ongoing shifts. Customers will buy new things from new sources and use them in new ways. Granularity is key. Shifts across many thousands of products will reshape the geometry of global trade.

Introduction

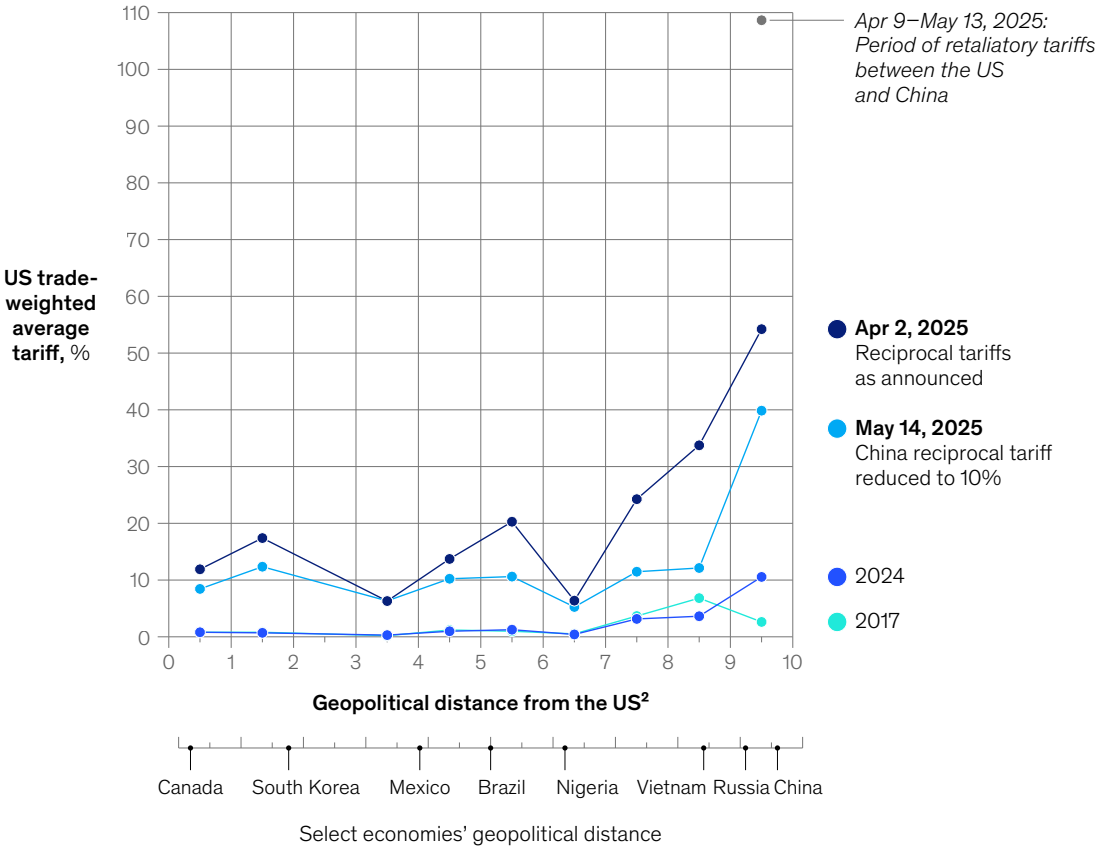
Tariffs have surged into the public spotlight. On April 2, 2025, the United States unveiled country-specific tariffs, defined by a formula based on goods trade deficits. Tariffs have substantially receded from those highs since and may continue to shift in the coming weeks and months as negotiations and court challenges unfold.

Yet substantial trade tensions between the United States and China could be here to stay.[1] When combined with prior policy measures, tariffs and geopolitics clearly correspond: economies that are more "geopolitically distant" from the United States, particularly China, tend to face the highest tariffs (Exhibit 1). If current settings persist, US imports may shift from China to countries that face lower tariffs and historically have been more geopolitically aligned with the United States.

Exhibit 1

Tariffs rise with geopolitical distance from the United States.

Trade-weighted tariff by economies' geopolitical distance from the US[1]



[1]Trade-weighted average within a "bin" of size 1 on the geopolitical distance scale (ie, 0–1, 1–2, etc). 2017 and 2024 data are historical actuals. 2025 tariffs are weighted using 2024 US product-level import values for each economy. Apr 2, 2025, tariffs consider the announced tariffs (and the Apr 11 clarification on exemptions), noting that these tariffs did not go into effect on Apr 2.
[2]Based on McKinsey Global Institute analysis of UN General Assembly voting records, 2005–22. This is used as a proxy for alignment on global issues but has some limitations. For discussion, see *Geopolitics and the geometry of global trade,* McKinsey Global Institute, 2024.
Source: US Census Bureau; McKinsey Global Institute Global Trade Dispute Database; McKinsey Global Institute analysis

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Even before 2025, the geometry of trade had been shifting along geopolitical lines. The average geopolitical distance of global goods started to compress beginning around 2018, particularly for the United States and China—evidence of so-called friendshoring. US imports from China fell by about 20 percent, or more than $100 billion, between 2018 and 2024, while total US imports rose by almost 30 percent over the same period.

Recent events may accelerate this realignment. Many US firms are urgently considering alternative sources of supply. Without a shift to different sources, prices may rise, and US companies and consumers might need to *reduce*—making do with fewer products or inputs—or *replace*, substituting one product for another sufficiently similar one. The higher the tariffs, the more significant the potential reduction or replacement.

An alternative is to *ramp up* existing US manufacturing capabilities (cars) and *rebuild* dormant ones (chips and ships). This change could happen in other countries, too. But it takes time, money, and know-how. And at least in the case of the United States, where wages are high, it may not be economically viable in some sectors.

Is it instead possible to *rearrange* trade in the nearer term, with the United States finding new suppliers and China finding new buyers? How feasible and costly is this path compared with reduction, replacement, and ramp-up? (see sidebar "Reduce, replace, ramp up, or rearrange? Changing trade patterns"). In this article, we introduce a "rearrangement ratio" to explore the potential for rearranging trade across a granular array of products and to examine how the geometry of global trade might shift.

1. Examining the potential to rearrange US–China trade

We define a rearrangement ratio to quantify how readily a country might shift its imports from one trading partner to others. For any given product, this ratio expresses the country's imports from that trading partner as a share of the total available remaining global export market. Higher rearrangement ratios imply greater difficulty sourcing a product from other countries (see technical appendix on page 25 "Understanding the rearrangement ratio").

The basic intuition is that if imports are small compared with all available global exports, rearrangement should be relatively easy. For example, the United States imports roughly $4 million worth of chocolate bars from China each year, while the available global export market—exports from all countries other than China and going to all countries other than the United States—is nearly $6 billion. Correspondingly, the rearrangement ratio is close to zero ($4 million of imports divided by a $6 billion available market). High tariffs on Chinese chocolates will not disrupt Americans' access to the confection.

But for ratios greater than 1.0, a country's imports exceed available global exports, making simple rearrangement impossible. For example, the United States sources about $3 billion in Christmas decorations from China. The total available global export market (that is, all Christmas decorations currently exported by economies other than China to countries other than the United States) is only $600 million. That leaves a huge gap, even if all available exports were diverted to the United States.

A high rearrangement ratio for US Christmas decorations might dampen the yuletide mood but does not pose major risks to lives or livelihoods. Disruptions in the supply of products such as semiconductors and lithium-ion batteries would have more meaningful consequences. Cutting back on consumption is not a good option. And some products, such as biologic medicines and semiconductor manufacturing equipment, require substantial investment and scarce know-how to produce, meaning that ramping up production is less feasible, at least in the short term.[2] For essential or complex products, if smooth rearrangement of trade is not possible, importers and end customers may need to absorb tariffs and pay more (see technical appendix on page 28 "Rearrangement ratio calculation: Approach and limitations").

Sidebar

Reduce, replace, ramp up, or rearrange? Changing trade patterns

An alliterative framework—*reduce, replace, ramp up, and rearrange*—helps structure thinking about what might happen to global trade patterns when things are in flux. This framework may characterize either a firm's actions or trade shifts at the country level.

Reduce—decreasing the consumption of products, by either consumers or firms. Increased tariffs impose a cost borne by some combination of players across the value chain. Consumption may drop, if prices rise. After the United States imposed tariffs on washing machines in 2018, their retail price jumped by 12 percent,[1] and, by some accounts, overall US sales remained depressed over the next few years.

Replace—substituting one product for another sufficiently similar one. If tariffs increase prices, or make a product harder to find, buyers may switch to an alternative. For example, China imposed retaliatory tariffs on its imports of sorghum—mainly used as animal feed—from the United States in 2018. This induced livestock farmers to substitute sorghum with alternative grains such as corn.

Ramp up—increasing the total amount of manufacturing or processing. Established manufacturers can build new factories, and new industries can develop (or old ones be revived). Ramp-up might occur in a low-tariff country seeking to export to the United States, or within the United States itself. For instance, when the United States imposed broad-based tariffs on China in 2018, production capacity in Vietnam ramped up, particularly linked to final assembly. US semiconductor production is currently ramping up, influenced by US tariffs together with domestic incentives and encouragement of foreign direct investment.

Rearrange—shifting who imports from whom, without changing total production or the overall trade balance. For example, when the United States imposed a 25 percent tariff on truck tires from China in 2018, US imports from China dropped by over 60 percent—but total US truck tire imports remained stable, as exports from other economies rearranged toward the United States.

[1] Aaron Flaaen, Ali Hortaçsu, and Felix Tintelnot, "The production relocation and price effects of US trade policy: The case of washing machines," *American Economic Review*, volume 110, number 7, July 2020.

Electronics and contract manufacturing are harder to rearrange

The average US China–rearrangement ratio is about 0.4, but the figure varies widely across sectors. It is lowest for pharmaceutical products (roughly 0.03), midrange (between 0.3 and 0.5) for electronics and textiles, and highest (more than 0.9) for "other manufacturing"—a catch-all category encompassing products from yoga mats to staplers to lawn furniture to knitting needles. These products are often contract (made-to-order) goods (Exhibit 2).

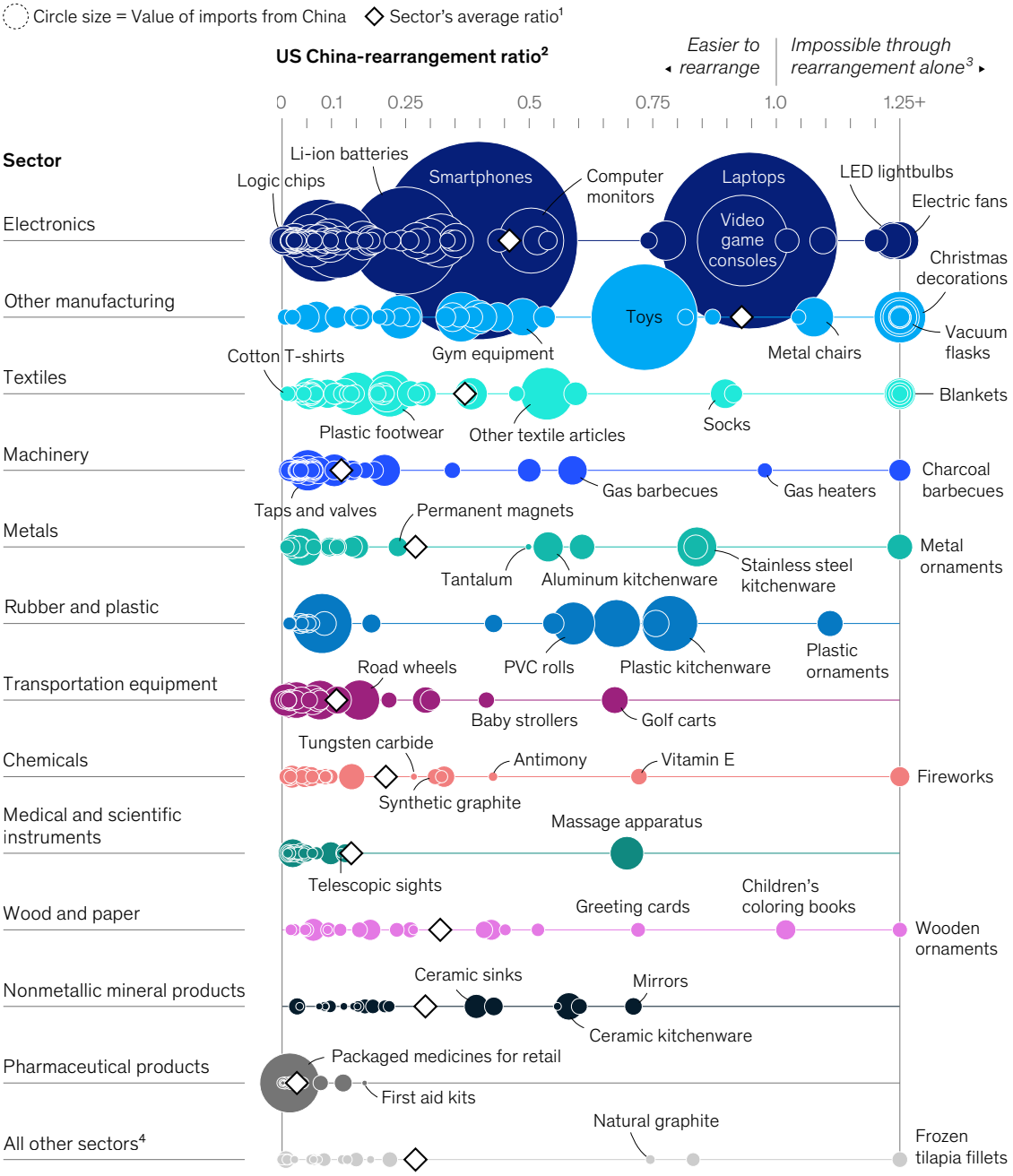
Sector averages mask large variation. Imports of consumer electronics are more difficult to rearrange than the intermediate components within them, like semiconductors. Almost all intermediate components have ratios lower than 0.3. In contrast, the most-imported consumer electronics have ratios ranging from about 0.3 to 1.0.

Consider smartphones, which are at the lower end of the consumer electronics range, with a ratio of about 0.4. While 80 percent of US imports come from China, manufacturers such as Vietnam and India currently export smartphones in sizable quantities to the non-US global market and so, in principle, could service the US market—if consumers were brand agnostic. For laptops, on the other hand, the rearrangement ratio is nearly 1.0. The United States imports more than $45 billion worth of laptops each year, more than 20 times more than it exports. China supplies about $35 billion of this total, roughly equal to all exports from Europe (the 27 member states of the European Union plus Norway, Switzerland, and the United Kingdom) and Asia to non-US markets.

Exhibit 2

Ease of rearrangement varies across products. Cotton T-shirts? Fairly easy. Fireworks? Impossible.

Rearrangement ratio for products imported by the US from China, by sector, 2023



Note: Sectors are ordered by total US import value from China in 2023. For clearer visualization, the top 20 US imported products from China for each sector are displayed; also included are all products with more than $250 million in US imports from China, as well as critical minerals with more than $50 million in US imports for which China is the largest US supplier. The displayed products represent more than 80% of US imports from China by value in 2023. Product labels are indicative abbreviations of the corresponding 6-digit Harmonized System description.
[1]Average weighted by US imports from China.
[2]US imports from China as a share of the "available" global export market for the US (ie, global exports from all economies excluding China that flow to economies other than the US).
[3]A ratio greater than 1 indicates that rearrangement alone cannot satisfy US import demand; however, demand could be met through ramped-up production, etc.
[4]Includes agriculture, energy resources, food and beverages, and minerals.
Source: UN Comtrade; McKinsey Global Institute analysis

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The range is also broad for textiles. In general, apparel such as T-shirts, jerseys, and trousers is exported by a wide range of economies, with rearrangement ratios generally less than 0.1. Indeed, China has been shifting away from this category and moved from representing nearly 45 percent of global apparel exports in 2010 to 25 percent in 2023. However, rearrangement ratios are greater than 0.5 for other products ranging from synthetic-fiber socks to face masks and surgical drapes.

Products within the other manufacturing category have rearrangement ratios from near zero to well past 1.0. Metal chairs and vacuum flasks are examples.

Of course, this is all approximation. The trousers shipped from Bangladesh may have a different cut, stitching, and fabric from those shipped from Italy. The cell phone manufactured in China may not be the same brand as the one made in South Korea. The coffee from Indonesia may taste more earthy, and that from Ethiopia, fruitier. But for our calculations, we treat products with the same classification, known as HS 6 codes, as interchangeable without considering quality factors or price.

US consumer goods imports from China are generally harder to rearrange than business inputs

Rearrangement is relatively easier for the roughly 35 percent of all US imports from China that have a rearrangement ratio less than 0.1 (Exhibit 3).[3] In practice, this means that to replace Chinese imports, US buyers would need to attract up to 10 percent of the currently available global export market. Even this would be a substantial undertaking. New input sources need to be found, product requirements specified, and contracts negotiated.

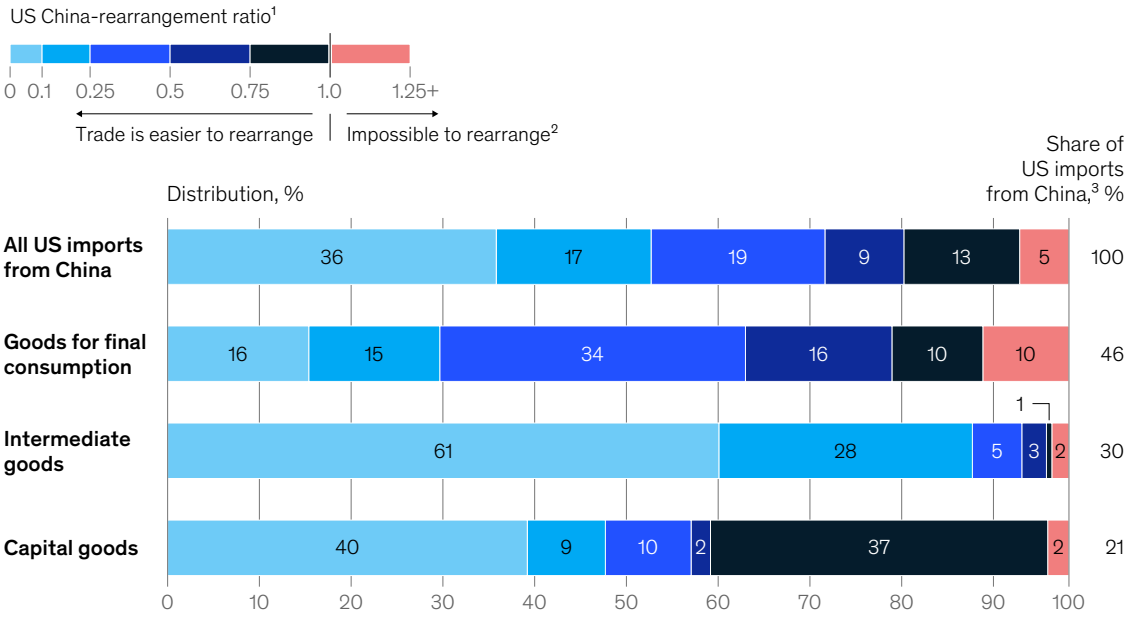
About 60 percent of intermediate goods—those used in production processes—fall into this easier-to-rearrange bucket. Examples include semiconductors and auto parts. An additional roughly 30 percent of intermediate goods have rearrangement ratios less than 0.25. The United States tends to import a large fraction of these types of products from economies other than mainland China, and an additional supply base that might be tapped exists.

About 60 percent of intermediate goods fall into the easier-to-rearrange bucket. Examples include semiconductors and auto parts.

Exhibit 3

Substituting US imports from China requires a major rearrangement of global trade.

Distribution of US imports from China by rearrangement ratio



Note: Figures may not sum to 100%, because of rounding.
[1]US imports from China as a share of the available global export market for US demand.
[2]A ratio greater than 1 indicates that rearrangement alone cannot satisfy US import demand; however, demand could be met through ramped-up production, etc.
[3]About 2% of US imports from China do not have a classification by Broad Economic Categories into goods for final consumption, intermediate goods, etc. As a result, these shares do not sum to 100%.
Source: UN Comtrade; UN Broad Economic Categories (5th revision); McKinsey Global Institute analysis

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On the other end of the spectrum, full rearrangement is currently impossible for the 5 percent of US imports that have a ratio greater than 1.0. These tend to be goods for final consumption—products that people buy for personal use—including consumer electronics as well as various forms of textiles and other manufacturing. In many cases, China has gained enormous cost advantages in producing these goods, making it both difficult and unappealing to compete.

Some specific critical intermediate products also have a ratio greater than 1.0. Examples include rare earth magnets, which are used in electric vehicles, wind turbines, and defense applications, as well as some chemicals used in pharmaceutical manufacturing.[4] In the case of rare earth magnets, more than 90 percent are produced in China, and China correspondingly supplies about 85 percent of US imports, so rearranging US imports to secure exclusively non-China sources of rare earth magnets is, at the moment, impossible. But supply chains that are less reliant on China are being ramped up.[5]

The fact that more critical minerals do not emerge with high rearrangement ratios may be surprising. China leads the refinement of a host of minerals from antimony to zinc, so one would think that finding alternative suppliers would be challenging. While true for some minerals, this is not always the case. Sometimes that's because US imports, for example magnesium, already skew to suppliers other than mainland China. In other cases, critical minerals are embodied in products the United States typically imports from economies other

than mainland China, such as gallium embodied in gallium-containing semiconductors, which the United States mainly imports from Taiwan, China.

For the remaining roughly 60 percent of US imports from China that have rearrangement ratios between 0.1 and 1.0, rearrangement is more difficult but not impossible. This middle category contains a large slug of both goods for final consumption as well as capital goods—the physical assets that support business production. By value, almost all of the capital goods with ratios higher than 0.5 are electronics, mostly laptops.[6] On the other hand, in the machinery and equipment sector, almost all capital goods have ratios under 0.25.

Overall, the average US China–rearrangement ratio has crept up slowly over time, nearly doubling since 1995 (see sidebar "Both US and China rearrangement ratios have increased gradually over time"). Most of that increase has come from an expanding fraction of products with rearrangement ratios near or greater than 1.0.

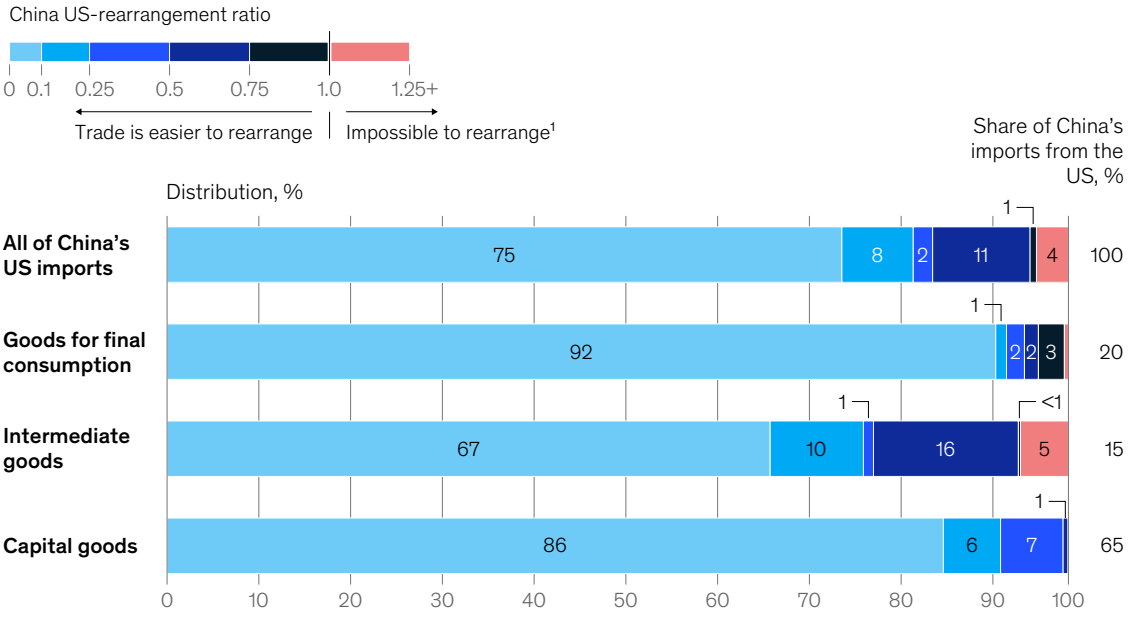
China's imports from the United States are easier to rearrange than US imports from China

A different picture emerges from the China US–rearrangement ratio, which considers how readily China might reconfigure its imports to non-US suppliers (Exhibit 4). This ratio is less than 0.1 for 75 percent of China's imports from the United States (recall that the equivalent share for the United States is about 35 percent), suggesting that China can more easily find alternate sources. This is partly because China imports a lot less from the United States than it exports to the United States—so there is less trade to rearrange. In addition, many of China's largest imports from the United States, such as crude oil, midsize cars, and corn, are widely exported by other countries, too.

Exhibit 4

Substituting China's US imports requires less trade rearrangement.

Distribution of China's imports from the US by rearrangement ratio



Note: Figures may not sum to 100%, because of rounding.
[1]A ratio greater than 1 indicates that rearrangement alone cannot satisfy China's import demand; however, demand could be met through ramped-up production, etc.
Source: UN Comtrade; UN Broad Economic Categories (5th revision); McKinsey Global Institute analysis

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Agriculture and chemicals make up nearly 70 percent of China's imports from the United States, with rearrangement ratios greater than 0.1. Soybeans—China's largest US import—have a rearrangement ratio of about 0.7, suggesting that reconfiguration is possible but challenging. The United States is the major global supplier of ethane, which China's petrochemical industry uses as a feedstock, leading to a rearrangement ratio greater than 1.0.

Overall, replacing China's imports from the United States through rearrangement is impossible for the 4 percent of imports that have a rearrangement ratio greater than 1—a similar share as the United States. However, while for the United States these products cover a range of lower-complexity consumer products, for China, they are limited to a few specific intermediate inputs.

2. Tracking the tariff cost curve

The current environment encourages rearrangement of trade between the United States and China. Differential tariffs across countries create a "tariff cost curve." For example, China still faces higher rates than other economies, even excluding those levied under the International Emergency Economic Powers Act, which are currently under legal debate (Exhibit 5).

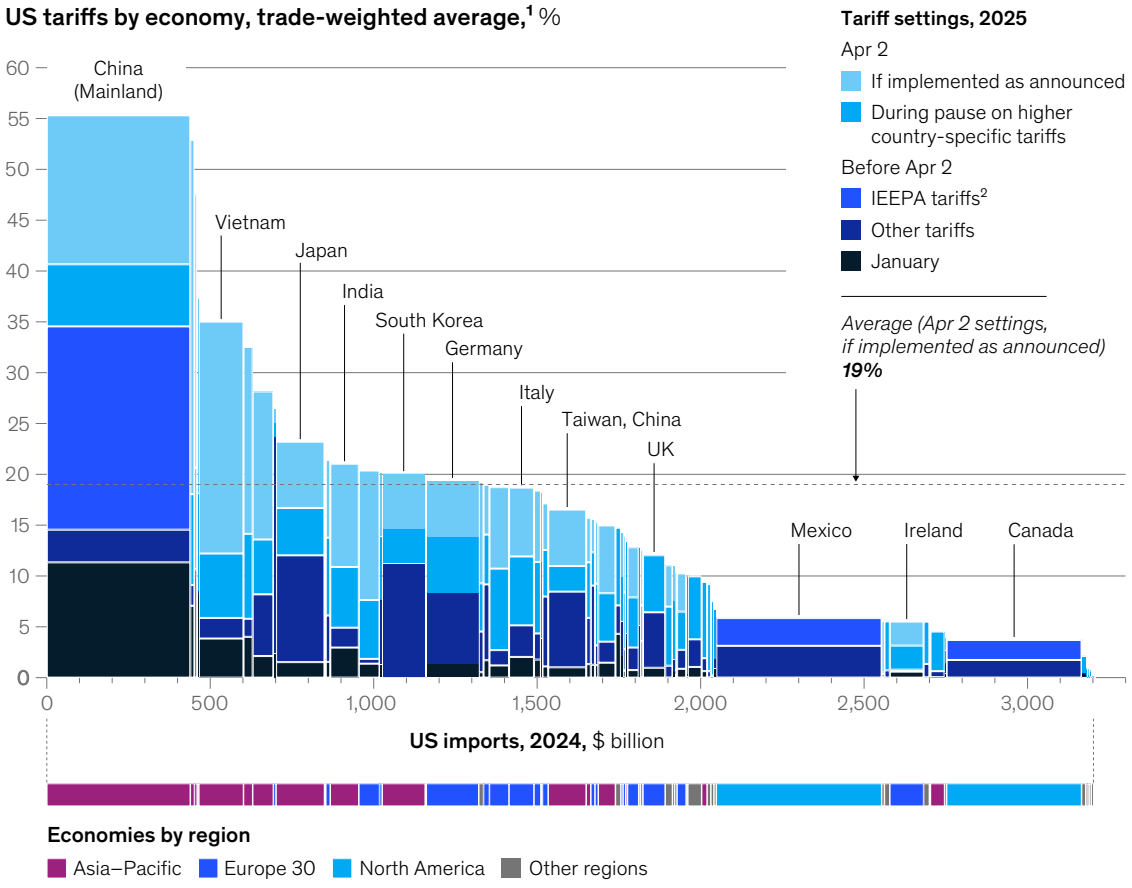
Of course, the tariff cost curve is not fixed—it has shifted significantly over the first months of 2025 and may continue to do so. But as long as some countries see higher tariffs (or other trade barriers) than others, it provides an impetus for US firms to consider shifting sourcing from higher-tariff economies on the left of the curve to lower-tariff countries on the right.

The cost curve varies by sector and by individual product, and it may continue to shift over time. But the same idea remains. As an example based on tariffs slated to start on August 12 (and subject to litigation and negotiation), if tariffs on cotton T-shirts from China were at 78 percent, a T-shirt exported to the United States from China would cost 1.78 times more than its pre-tariff price. A T-shirt imported from Guatemala would cost only 1.1 times more, reflecting Guatemala's 10 percent tariff (also subject to legal proceedings) and preexisting trade agreements. As long as the all-in cost of the Guatemalan version is less than 1.6 times China's—accounting for any differences in factors including local production costs, transportation,[7] and profit margins—it would be less expensive. This same logic applies under different potential future tariff trajectories, even if specific numbers shift.

China's imports from the United States are easier to rearrange than US imports from China. Many of China's largest imports from the United States are widely exported by other countries.

Exhibit 5

Recent announcements create a US tariff cost curve.



Note: For clearer visualization, excludes economies representing less than 0.1% of US goods imports in 2024. The included economies represent more than 98% of US goods imports in 2024.
[1]Tariffs are weighted by 2024 US product-level import values. Tariffs include all US tariff policy settings up to and including the date listed above. Apr 2, 2025, settings include the Apr 11 clarification on exemptions. Note that trade-weighted tariffs may differ from the country-specific tariffs announced on Apr 2; this is due to differences in trade composition, product exemptions, and other US tariff policy measures.
[2]International Emergency Economic Powers Act. Implementation of tariffs under this act is currently under legal challenge.
Source: US Census Bureau; McKinsey Global Institute Global Trade Dispute Database; McKinsey Global Institute analysis

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Sidebar

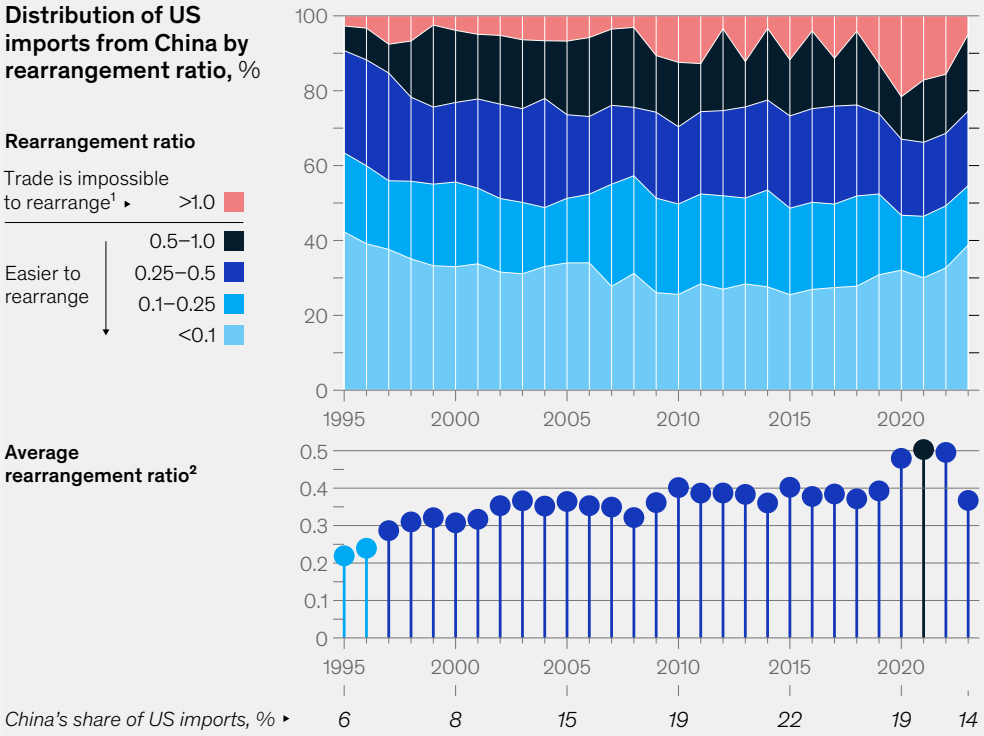
Both US and China rearrangement ratios have increased gradually over time

The average US China–rearrangement ratio has crept up slowly over time (Exhibit 1). It hit a pandemic-era peak of 0.5 in 2021, with rising US imports from China of previously less-in-demand goods from masks to standing desks. The gradual change in US China–rearrangement ratios belies an underlying shift in the most difficult-to-replace products. Much of this shift has been toward consumer electronics and household appliances. In 2000, China exported 20 percent of the world's microwave ovens and 1 percent of its laptops. Today it exports 70 percent of both.

The average China US–rearrangement ratio has doubled over the past 30 years, from just over 0.1 to about 0.2. It hit a peak in the mid-2010s but then declined around the time of the US–China trade dispute, driven by shifts in China's imports of some products with higher rearrangement ratios—such as soybeans and airplanes—to alternate sources.

Exhibit 1a

Rearranging US imports has become progressively more difficult.



Note: Due to inconsistent reporting of re-exports in the historical data, this analysis uses economies' export and import data to create a consistent time series. This creates a slight difference between the 2023 figures reported here and other 2023 figures in this report, which exclude re-exports.
[1]A ratio greater than 1 indicates that rearrangement alone cannot satisfy US import demand; however, demand could be met through ramped-up production, etc.
[2]Value-weighted average rearrangement ratio across all products imported by the US from China.
Source: UN Comtrade; McKinsey Global Institute analysis

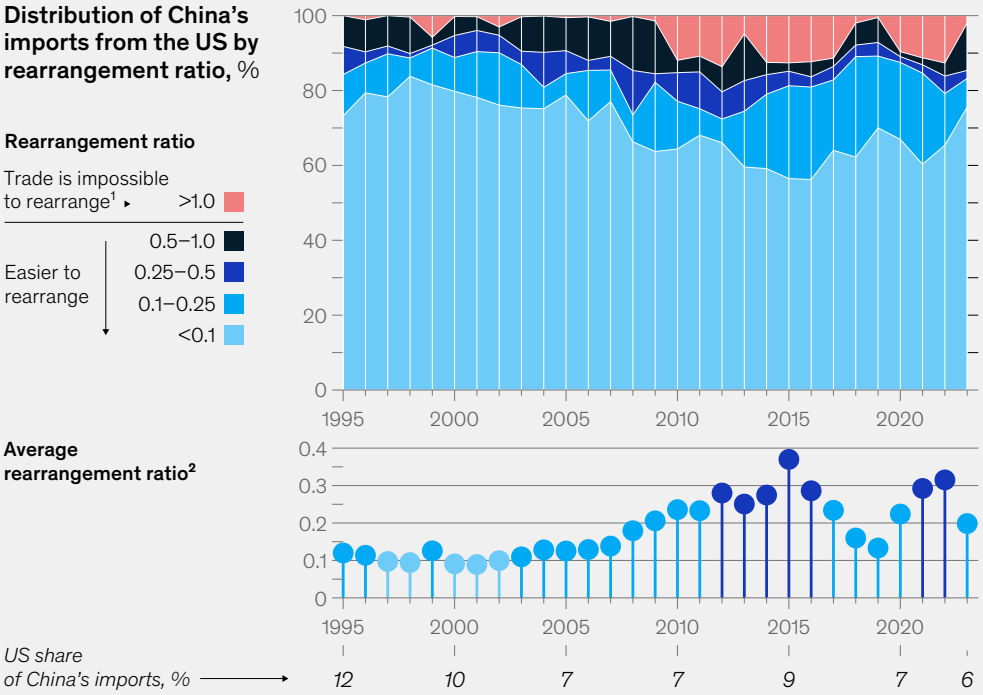
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Sidebar

Both US and China rearrangement ratios have increased gradually over time (continued)

Exhibit 1b

China's imports from the United States would be easier to rearrange.



Note: Due to inconsistent reporting of re-exports in the historical data, this analysis uses economies' export and import data to create a consistent time series. This creates a slight difference between the 2023 figures reported here and other 2023 figures in this report, which exclude re-exports.
[1]A ratio greater than 1 indicates that rearrangement alone cannot satisfy China's import demand; however, demand could be met through ramped-up production, etc.
[2]Value-weighted average rearrangement ratio across all products imported by China from the US.
Source: UN Comtrade; McKinsey Global Institute analysis

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Currently announced tariffs are higher in sectors where rearrangement is harder

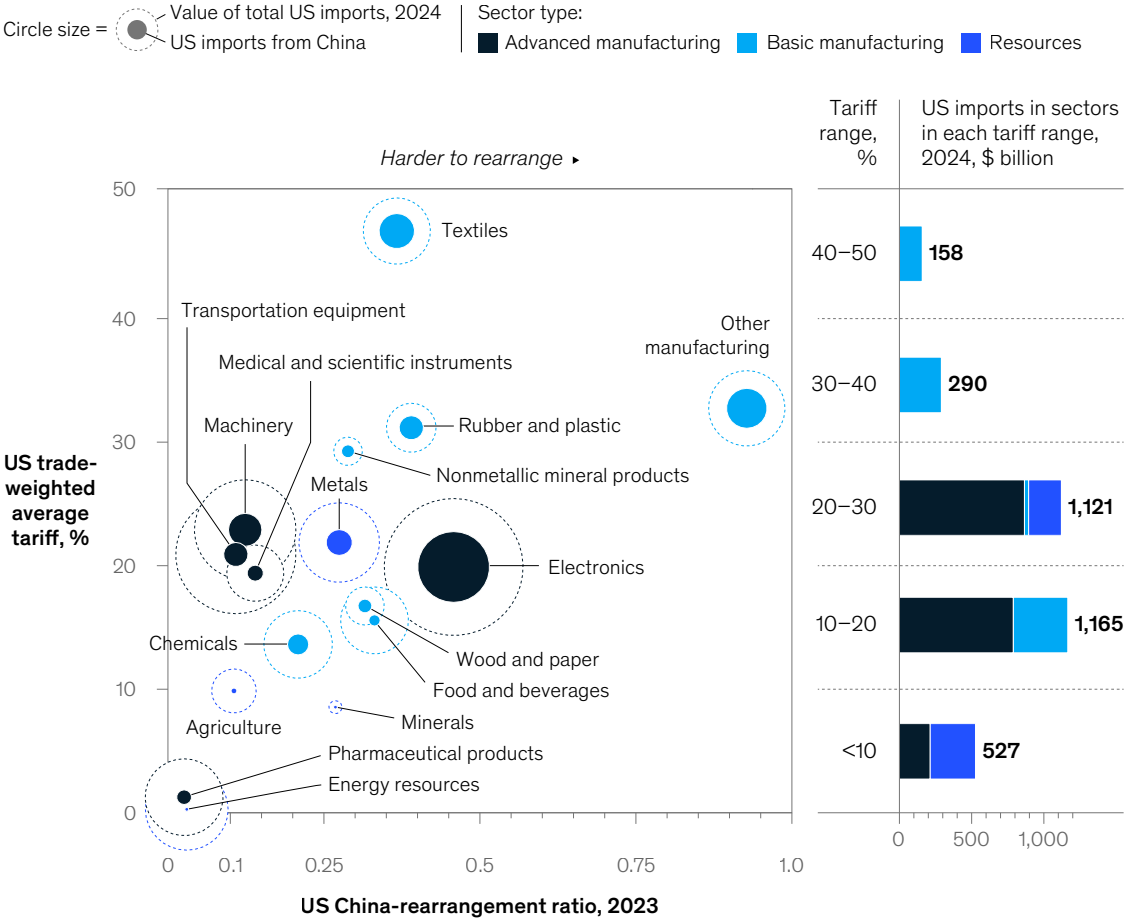
US tariff rates announced on April 2, 2025, vary significantly by sector, even though they largely were set on a country-specific basis. Basic manufacturing sectors, including textiles, rubber and plastic, and other manufacturing, tend to face higher tariffs. Many major exporters of these types of products are low- and middle-income countries that tend to run trade surpluses with the United States and so had higher country-specific tariffs.

Basic manufacturing sectors also tend to have higher US China–rearrangement ratios because the United States is often an outsize buyer, and China is an outsize seller. As a result, the tariff pressure to rearrange trade may be highest in those sectors where rearrangement is likely to be the most challenging (Exhibit 6).

Exhibit 6

Hard-to-rearrange sectors face higher tariffs.

US tariffs as announced on Apr 2, 2025, and rearrangement ratio[1]



[1]Tariffs are weighted by 2024 US product-level import values from each economy. Tariffs include all US tariff policy settings, up to and including the announcement of Apr 2, 2025. Apr 2 settings include the Apr 11 clarification on exemptions.
Source: US Census Bureau; UN Comtrade; McKinsey Global Institute Global Trade Dispute Database; McKinsey Global Institute analysis

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Most advanced manufacturing goods, including electronics and machinery, are exported in meaningful amounts by China as well as a range of advanced—and geopolitically closer—economies in Europe and Asia that do not have such large trade imbalances with the United States. Therefore, advanced manufacturing sectors would see relatively smaller tariffs, if these are set based on goods trade deficits, and also may be relatively easier to rearrange. Consumer electronics is the biggest exception.[8]

3. However rearrangement plays out, broad patterns hold

Most countries would feel knock-on effects if the United States and China rearranged trade away from each other. Some would export more to the United States, making up for previous supply from China. This could leave a supply vacuum elsewhere, which China's exports might fill.

For example, the United States receives about 70 percent of its $20 billion in lithium-ion battery imports from China. Poland is the second-largest exporter of lithium-ion batteries, accounting for about $12 billion annually. Roughly 80 percent of this total remains within Europe. Were Poland to rearrange exports to the United States, this would go a long way toward meeting US demand. But there would be a shortfall in European supply, which might be met by an increase in the number of batteries shipped from China to Europe.

Rearrangement could occur in many ways. We explored a range of simulations for how US imports from China might rearrange (Exhibit 7). These are not real-world scenarios—for example, each assumes the same approach for rearrangement across all products—but together they cover a broad landscape of potential approaches (see technical appendix on page 29 "Trade rearrangement approaches: Definitions and limitations"). And despite this variety, some striking common patterns emerge:

— Europe plays a key role in trade rearrangement, both as exporter and importer.

— Southeast Asia plays a significant part making up for China's electronics exports.

— Canada and Mexico see a surprisingly small impact on exports from rearrangement.

— The United States may turn to its own export supply, disrupting key trade partners.

— Product shortfalls and excess supply may rattle consumer electronics and other sectors.

These patterns, such as how Europe's exports and imports evolve, are constant across a range of rearrangement possibilities. Initially, US buyers need to decide how they source alternative supply, which affects the resulting global rearrangement patterns. For example, do they look to the lowest-tariff countries first? Do they seek stability by looking to countries that are closest geopolitically to the United States? Do they prioritize rearrangement simplicity by approaching sellers with the largest potential supply in order to minimize the number of new commercial relationships they need to form (see sidebar "Rearranging US imports requires sourcing from many economies")?

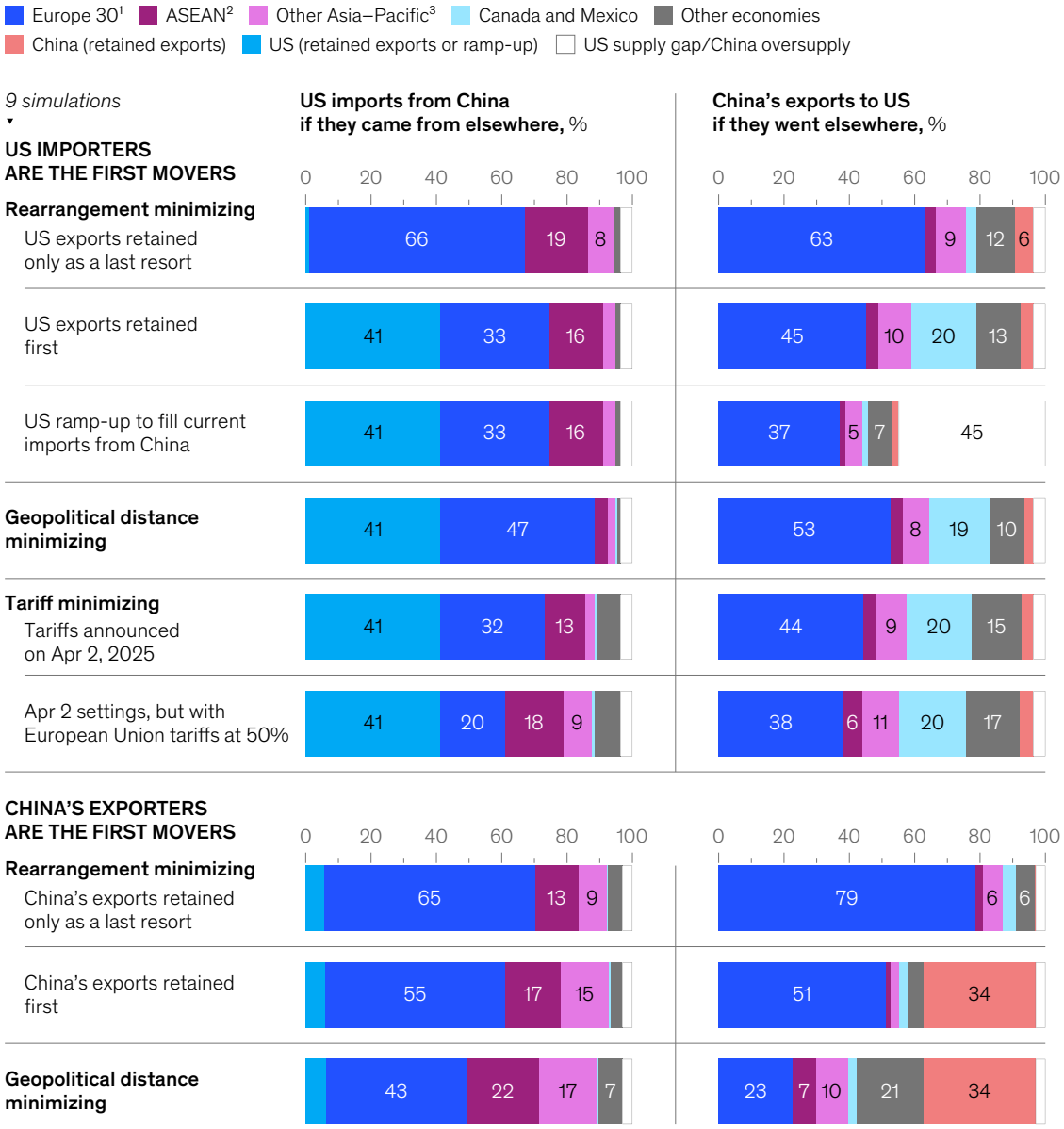
There is also a range of possibilities for current US exports. To what degree might they help replace imports from China, as either a first choice or a last resort? Could US producers manage to boost production and thereby reduce the country's trade deficit by filling the gap from Chinese imports at home while continuing to sell abroad? Of course, this last option requires ramp-up of production and so goes beyond pure rearrangement.

US importers may not be the first to act: China's exporters might make the first move by shifting from the United States to other markets. In the battery example above, this would entail China turning from the United States and selling more to Europe. Polish manufacturers could then face relative oversupply in their local regional market—but also increased demand in the United States, which would receive fewer imports from China. Polish manufacturers could therefore elect to sell more to the United States.

Exhibit 7

Europe emerges at the center of trade rearrangement.

9 ways that $440 billion of US imports from China could be rearranged



[1]European Union economies plus Norway, Switzerland, and the UK.
[2]Association of Southeast Asian Nations.
[3]India and Asia–Pacific economies that are not otherwise included in ASEAN and China.
Source: UN Comtrade; McKinsey Global Institute Global Trade Dispute Database; McKinsey Global Institute analysis

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Sidebar

Rearranging US imports requires sourcing from many economies

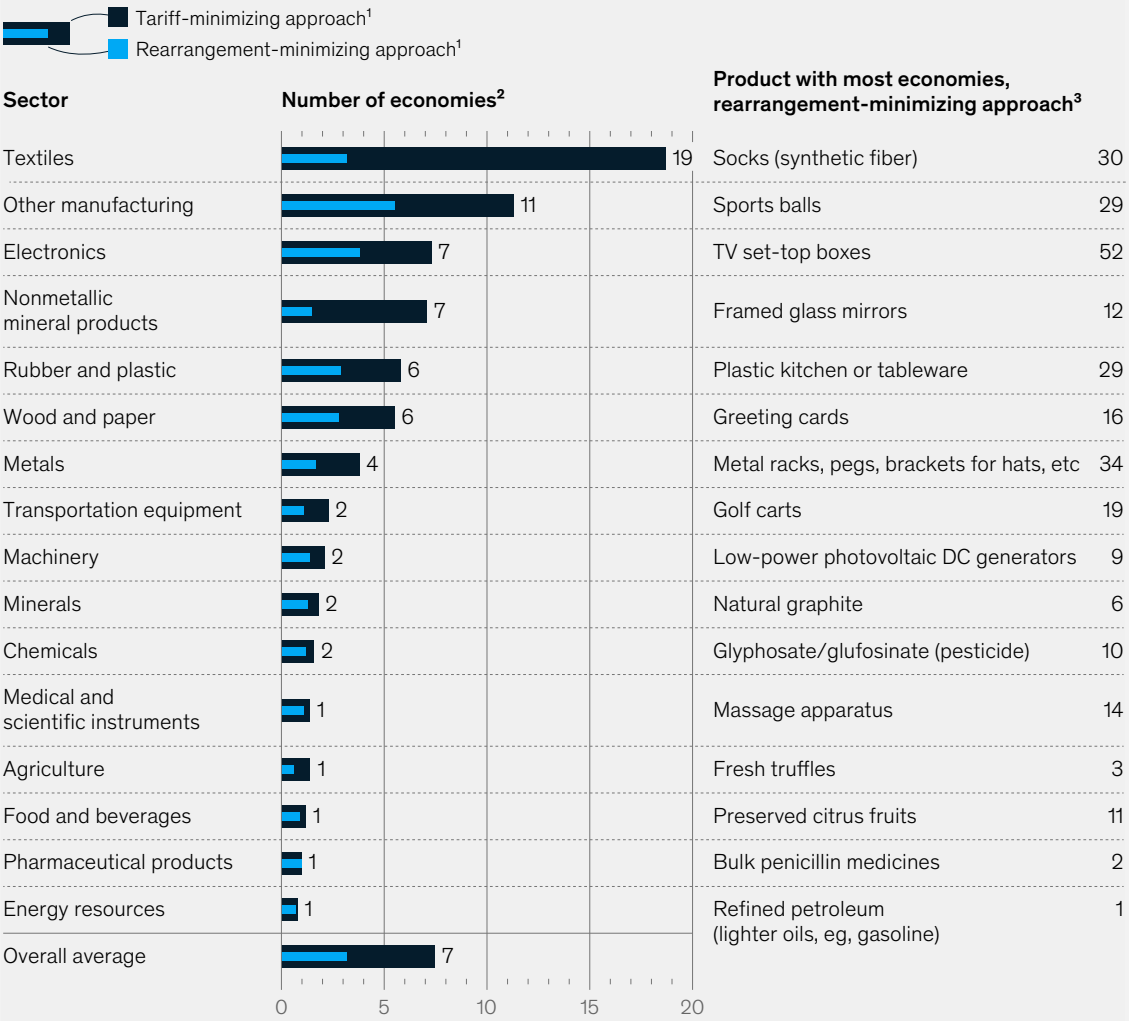
The practical challenge of shifting sourcing to the lowest-tariff supplier may be substantial. For textiles, it means securing, on average, the entire export market from 19 different countries to replace China's supply (exhibit). There is a clear first-mover advantage for importers. Across products, first movers face either no tariff (if sourcing from the United States) or, on average, a 5 percent tariff. The last movers, on the other hand, face a 22 percent tariff on average—assuming tariff settings as announced on April 2, 2025.

Some of the challenges may be addressed by a rearrangement-minimizing approach, which replaces US supply from China with the fewest possible exporters. Under this method, three exporting countries, on average, may replace US imports from China for each product. However, there is substantial variation by product: for example, for synthetic-fiber socks, exports from 30 countries would be needed. Moreover, there is a potential trade-off between tariff rate and volume—the countries with the largest available supply of a given product on average have tariffs that are more than 20 percentage points higher than those on the lowest-tariff suppliers.

Exhibit

A 'tariff-minimizing' approach entails more rearrangement.

Number of economies used to source incremental US supply for each product, sector average



[1]Follows the "US last" rearrangement-minimizing approach (simulation 1) and the tariff-minimizing approach with US tariffs as announced on April 2, 2025 (simulation 5). See sidebar "Trade rearrangement approaches: Definitions and limitations" for details.
[2]Simple average across all products for which the total US import value in 2023 was more than $100 million. These represent more than 95% of US imports by value.
[3]Excludes products for which US demand cannot be met through reconfiguration, which require trade diversion from all exporting economies in the modeling, and excludes catch-all products with "not elsewhere specified" descriptions. Product labels are indicative abbreviations of the Harmonized System 6-digit code for the relevant products.
Source: UN Comtrade; McKinsey Global Institute Global Trade Dispute Database; McKinsey Global Institute analysis

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Europe emerges as the focal point of rearrangement as both exporter to the United States and importer from China

Across simulations, Europe[9] sees substantial changes as both importer and exporter. It is a significant producer of many of the products the United States buys from China; on average, Europe supplies 55 percent of the available global export market for products that the United States imports from China. So it might rearrange its exports toward the United States.

Across most of our simulations, European exports to the United States substitute about 30 to 65 percent of China's current exports, and Europe's US exports go up meaningfully, even in our simulation with a high US tariff on European imports. This would increase Europe's trade surplus with the United States—unless it simultaneously increased its purchases from the United States—with potential implications for future tariffs and trade negotiations.

At the same time, Europe's imports from China also stand to go up substantially, at least given current tariff and other policy settings. In most simulations, Europe absorbs between 35 and 55 percent of what China is currently sending to the United States. Large supply chain vacuums might occur in Europe, whose exports under the simulations are rearranged from being mostly intra-European toward supplying the US market. Since European and US imports share similar profiles, it is a natural replacement market for China's exports if they do not face significant trade barriers.

Europe is central in the trade rearrangement across most sectors (Exhibit 8). The three largest US import sectors from China—electronics, other manufacturing, and textiles—contribute most to overall rearrangement. Across simulations, Europe increases its US exports in these areas. By value, electronics rise the most. Exports of some products in other sectors also rise substantially. Toys are an example: American kids may play with Czech and German toys rather than toys from China, and European kids may play with Chinese toys instead of toys from Germany and the Czech Republic.

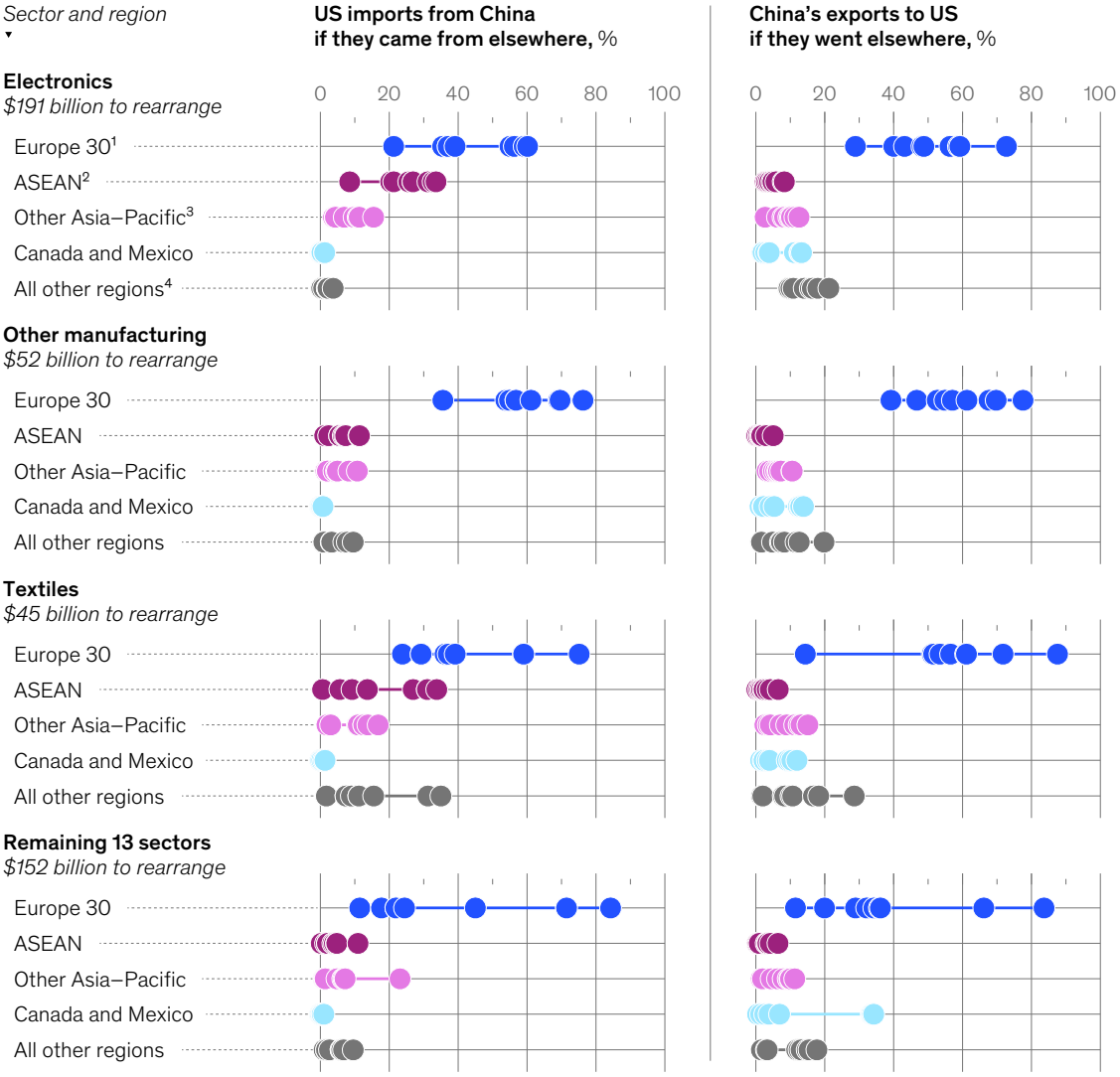
This all underscores the potential deepening trade interdependence between Europe and China. Another stark effect: China would overtake Germany as the largest exporter to the United Kingdom and the Netherlands.

Exhibit 8

Europe's central role plays out across sectors.

Regional share of rearranged US–China trade

●–● Range of percentage shares (each circle = 1 of 9 simulations)



[1]European Union economies plus Norway, Switzerland, and the UK.
[2]Association of Southeast Asian Nations.
[3]India and Asia–Pacific economies that are not otherwise included in ASEAN and China.
[4]Excluding the US and China.
Source: UN Comtrade; McKinsey Global Institute Global Trade Dispute Database; McKinsey Global Institute analysis

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ASEAN plays a significant role in making up for China's electronics exports

Association of Southeast Asian Nations (ASEAN) member states also play a meaningful role in making up for China's exports, but it is less broad based and more sector specific than in Europe. Across the rearrangement approaches we explore, ASEAN exports to the United States replace up to about 20 percent of China's current exports.

Much of this effect comes from a single sector, electronics, with most of the rise driven by exports from Vietnam. Indeed, Vietnam accounts for more than 80 percent of the increase in ASEAN's electronics exports in most of the simulations we run, becoming the largest supplier of US laptop and smartphone imports as a result.

ASEAN members could also replace a substantial share of China's textile exports to the United States, but this is more dependent on the specifics of rearrangement. For example, with tariffs as outlined on April 2, 2025—which are typically higher for ASEAN members—a tariff-minimizing approach results in US textile imports coming more from economies such as Morocco and Türkiye. ASEAN members substitute only about 5 percent of US imports from China. Conversely, under some rearrangement-minimizing approaches, this share rises to nearly 40 percent.

Rearrangement may leave some big US trading partners' exports largely untouched

Not everyone would export more to the United States. On the surface, one of the more surprising findings from rearrangement is that the country's neighbors and biggest trading partners, Canada and Mexico, do not see much of a lift in exports to the United States. This reflects similar export profiles and the already high integration of North American manufacturing. The vast majority of Canada's and Mexico's goods exports—more than 80 percent—already go to the United States, leaving less untapped rearrangement potential than in other regions. This is reflected at the sector level, too. For example, Mexico experiences relatively small textile gains across simulations, as more than 90 percent of these exports already go to the United States.

Exports to the United States from Japan and South Korea also typically rise very little, by only about 5 percent, across most simulations. These economies lean toward exporting complex intermediates rather than the final goods that the United States typically buys from China. For example, while China currently exports about $120 billion in laptops to the world, Japan exports less than $400 million worth.

Exports from a range of other locations also remain relatively constant. For example, Sub-Saharan Africa, Latin America, and the Middle East and North Africa each substitute about 1 percent of China's exports across simulations. While not a big deal for an economy the size of the United States, such increases can be meaningful for a smaller country's exports. In the US rearrangement-minimizing simulations, for example, about 40 countries experience increases in their exports to the United States of more than 10 percent. About half of these countries are in Europe, and the remainder in the developing world. In the tariff-minimizing simulations, it's broader based. About 70 countries—with the additional ones mostly in Africa, Latin America, and the grouping of countries in "Other Europe and Central Asia"[10]—register increases greater than 10 percent in their exports to the United States.

The United States may keep more of its exports home to replace Chinese imports

A significant swing factor influencing rearrangement dynamics is the degree to which current United States exports make up for imports from China.

In 2023, the United States imported roughly $440 billion in goods from China and exported $1.7 trillion across the world.[11] Product-level analysis suggests that the United States might draw on about $180 billion worth of those existing exports before seeking potentially more expensive overseas alternatives. As a result, the country may itself be the largest source of goods to make up for those that are no longer imported from China.

One example: in 2023, the United States imported about $2 billion in electrical cables from China, but

it also exported more than $2 billion worth. It could just keep that $2 billion in cables home. In reality, it is unlikely that a cable from China could be fully swapped for a US-made one.[12] This would require retooling of existing manufacturing.

How the United States adjusts affects other economies. If the United States does not draw on its existing exports then, as discussed above, European exports to the United States may increase substantially, given their similar profile. In contrast, if it does draw on its own potential exports to meet domestic demand, then Canada and Mexico might experience notable increases in their imports from China, as China comes to supply goods that Canada and Mexico currently buy from the United States. Latin America exhibits a similar pattern.

If the United States ramps up production to provide for its own domestic needs while continuing to export, then the shift is even more extreme for China. In most simulations, China's surplus exports amount to about $20 billion that cannot be rearranged to other partners. In the US ramp-up case, this figure rises to almost $200 billion, meaning an extra $180 billion of Chinese exports that don't have a home. This implies an equivalent decrease in China's trade surplus, and a reduction in the US trade deficit.

The United States may face a $100 billion supply gap, and global markets a similar oversupply, on goods like laptops and smartphones

Rearranging trade at a large scale poses significant challenges. New suppliers need to be located, manufacturing specifications met, and multiyear contracts negotiated. For example, automotive inputs contracts, including parts for original equipment manufacturers and critical mineral deals for battery production, have had durations of about three to five years, and sometimes more. Generally, for contracts with an average duration of four years, roughly one-quarter would likely come up for negotiation each year. So, for a product with a ratio of 0.25, full rearrangement within 12 months would require shifting roughly all contracts up for negotiation that year—a tall order.

In the case that, for each product, the United States secures only 25 percent of the available global export market, it would see an import shortfall of more than $100 billion, and other countries throughout the world might see surplus exports from China of the same amount. Products at the center of modern life including laptops, computer monitors, and smartphones may be affected, as might a range of common items from toys to kitchenware to golf carts to quilts (Exhibit 9). Many of these may be nice, or even important, to have but aren't necessarily essential. In that case, one response from US customers may be to turn their household budgets away from the now higher-priced imports.

As an example, if the United States no longer imported laptops from China, this could result in a supply shortfall of nearly $26 billion. Consumers and businesses may experience higher prices as a result. The global, non-US market could be oversupplied by the same amount, equivalent to about 25 percent of additional supply. This may lead to lower prices outside the United States, bringing potential benefits for consumers, but challenges for manufacturers.

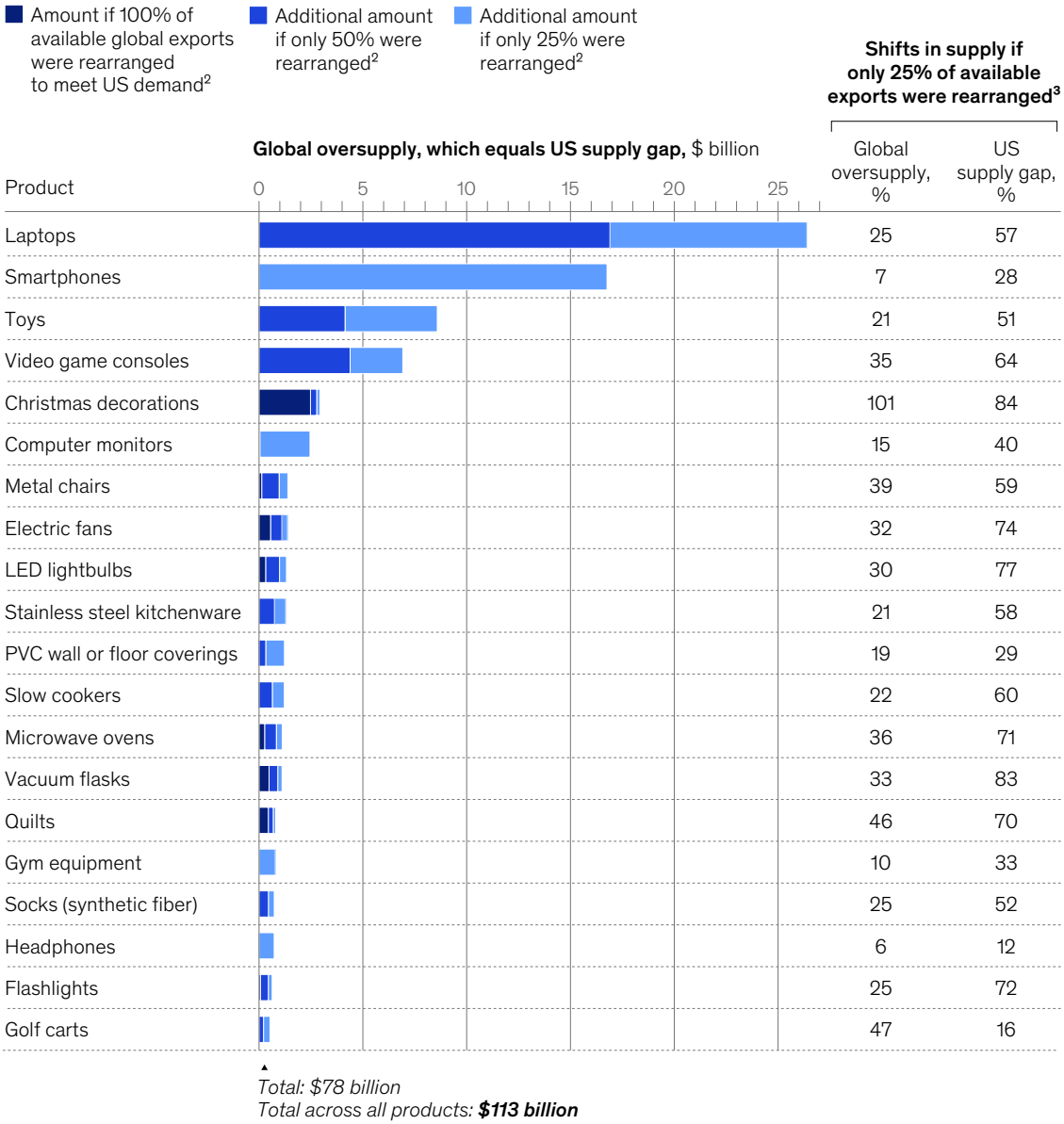
But rearrangement would not materially affect all of China's major exports. Many do not go to the United States in significant quantities, in part because of historical US tariff policies.[13] For example, the United States imports between $15 billion and $35 billion annually in the categories of electric vehicles (EVs), logic chips, solar panels, and steel mill products—all of which are also some of China's largest exports. However, China supplies less than 3 percent of US imports for all of these.

Of course, shortages and surpluses aren't guaranteed. This analysis looks at the geometry of global trade and its potential to accommodate rearrangement but not changes in capacity and domestic consumption that will surely follow.

Exhibit 9

Global supply of some products could increase substantially, depending on rearrangement.

Global oversupply of select products[1]



Note: Product labels are indicative abbreviations of the Harmonized System 6-digit code for the relevant products.
[1]Products are selected from those with more than $500 million of global oversupply, if 25% of the available global export market rearranges to meet US demand.
[2]A figure of 100% indicates that the entire global export market can rearrange to meet US demand if US supply from China falls to zero, 50% indicates that up to half of the global export market can rearrange to meet US demand for any product, and so on. If the resulting incremental exports to the US are insufficient to meet 2023 US import demand, this results in a hypothesized US supply shortfall and an equivalent export excess by China of that product to the global (non-US) market.
[3]Global oversupply and the US supply gap are expressed as shares of the 2023 total import value for the global (non-US) and US import markets, respectively.
Source: UN Comtrade; McKinsey Global Institute analysis

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4. Reduce, replace, ramp up—or rearrange?

Multinational corporations may have to navigate an environment in which the geopolitical distance of US trade shortens, the China–Europe trading relationship deepens, and product-level surpluses and shortages emerge across countries, at least in the near term. Smaller firms and national champions will have an opportunity to enter an upended market in new ways.

That said, the pain of tariffs may be substantial for some firms, and uncertainty is high for everyone, with readings of policy uncertainty hitting near-record levels. Changing course requires up-front investment, and so some firms are waiting to act in hopes that conditions become more certain. But at the same time, acting first can have advantages. Firms may consider actions across *reduction, replacement, ramping up,* and *rearrangement* to respond to potential trade disruptions, bearing in mind that their upstream suppliers, downstream customers, and competitors all face the same range of choices. Of course, governments themselves may also shift the trade landscape over time. Questions to consider include the following:

— *Reduce.* Where the United States is hit by shortages of certain products, reduction may be the only choice in the immediate term. It can also be a strategic option. Can needs be reduced through efficiency improvements? Is it the right moment to sell less but adjust the pricing approach or target customer segment? Should this be a trigger to dial down certain business lines?

— *Replace.* This can also be an immediate option, and some substitutions are straightforward with little sacrifice on quality. But unplanned, last-minute replacement can be costly and suboptimal. Which of the products that firms buy or sell should have a replacement contingency plan? More strategically, can products or production processes be updated so replacement becomes structurally easier? As with reduction, is there an opportunity to shift pricing, segment, or business focus?

— *Ramp up.* In certain scenarios, the United States may face ongoing supply disruptions, creating opportunities for suppliers both domestically and overseas. What conditions need to be met in order for the economics of ramping up to work, considering up-front costs, steady-state economics, market opportunity, tariffs, and level of certainty about all these factors? Are there contingency plans for ramping up or for working with a supplier who can, including a planned trigger for action? Might preemptive ramping up give first-mover advantages?

— *Rearrange.* Since rearrangement ripples across the globe, it raises questions from many interconnected perspectives, including those of US firms importing, foreign firms looking to export either more or less to the United States, and firms that don't interact with the United States directly but operate in markets where other firms do. Competition among US firms may be high to secure supply from the lowest-tariff countries—with specifics depending on the future trajectory of tariffs—or from domestic alternatives. How large is the first-mover advantage, and how can buyers ensure that they are better placed than competitors in any forthcoming rearrangement?

  Some exporters, in countries facing lower tariffs, may be looking to double down on the US market. How can such firms position themselves, and in which specific product and customer segments? Other firms that currently ship to the United States may be seeking to reorient to places where they face lower tariffs or less uncertainty. Where are these non-US markets, will they face intensified sales competition, and how can businesses prepare?

  Even firms that do not buy from or sell to the United States may explore whether they will feel ripple effects. Certain markets may experience surpluses in some products if Chinese or other exporters pivot to them from the United States. In this case, is there an opportunity to negotiate lower prices? Are there products where supply may be reduced if their sellers have newly turned toward the United States? How can buyers prepare for this form of disruption, and how can sellers take advantage of potential gaps?

  When thinking about these questions, firms need to consider approaches that are resilient to—potentially ongoing—shifts in policy. If rearrangement is sufficiently easy and the required sunk cost is low, it may be preferable to the three alternatives. On the other hand, the highest rearrangement ratios signal which products carry the strongest incentives for a different form of adaptation.

No matter the balance between rearranging—or reducing, replacing, and ramping up—the current disruption suggests that customers will buy new things, from new sources, and use them in new ways. US consumers may switch brands more frequently and try new products from home or abroad. Producers may take margin or market share in the domestic market. Global sellers could get a chance to crack the US consumer market. Chinese producers may have greater opportunities to expand into markets that are currently served by others. Flexible payments systems, diversified currency baskets, and liquidity measures might grow more important. Strategy certainly will.

Specifics matter. Shifts may look very different across the many thousands of products that make up the global web of trade. When making strategic decisions for their organizations, leaders should examine fine-grained product market dynamics to understand where there may be surpluses or shortages, price sensitivity or rigidity, and granular pockets of decline or growth.

Trade rearrangement promises to reshape the geometry of global trade, setting the backdrop as firms build resilience in a reordering world.

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Understanding the rearrangement ratio

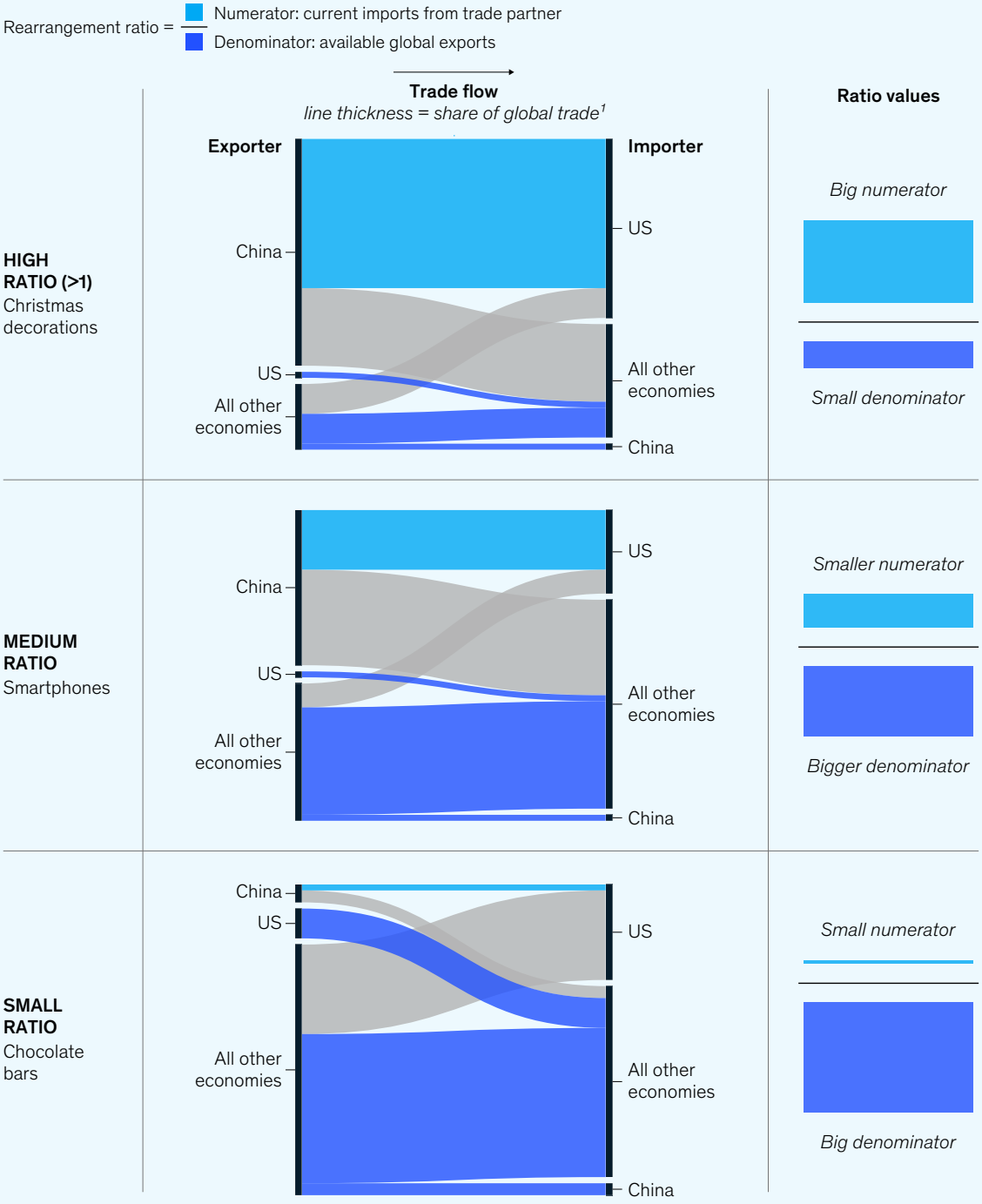
If trade flows are disrupted, rearrangement of existing trade may enable an importing country to meet demand for products without any production ramp-up. The rearrangement ratio quantifies at the product level how readily this might occur if imports to country A from country B were interrupted. The numerator contains country A's total imports from country B. The denominator comprises all total available global exports that might replace these imports—that is, all exports shipped throughout the rest of the world, as well as all of country A's own exports (which might instead serve domestic demand currently met by imports from country B). A few schematic examples illustrate the idea (Exhibit 1).

Rearrangement ratios can be calculated for trade between any two countries. Exhibit 2 shows average rearrangement ratios for seven major trading economies. The China Brazil-rearrangement ratio is highest, dominated by China's massive $40 billion imports of soybeans from Brazil. This dwarfs the potentially available soy export market, which is about $20 billion, because other exporters already ship most of their soybeans to China. Beyond US–China ratios, other rearrangement ratios greater than 0.2 are China–India and India–US. Often, these midrange ratios are driven by acute points of reliance for specific products. For India's exports to China, for example, the higher ratio is driven by products like castor oil. India is the largest global supplier of castor oil, and castor oil is one of China's largest imports from India, used mainly as a building block in its chemical industry. China gets more than 99 percent of its castor oil imports from India, and the rearrangement ratio for this product is correspondingly high.

Exhibit 1

Defining the ‘rearrangement ratio.’

Illustrative examples of rearrangement ratios for US imports from China



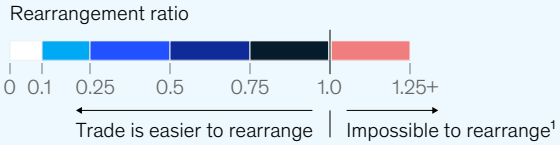
[1]Line thicknesses are representative of global trade data but have been adjusted for illustrative purposes.
Source: UN Comtrade; McKinsey Global Institute analysis

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Exhibit 2

The US China-rearrangement ratio is the second-highest among large economies.

Rearrangement ratios for select economies, weighted by total trade flows from exporter to importer



Importer \ Exporter	United States	China (Mainland)	Germany	Japan	India	United Kingdom	Brazil
United States		0.41	0.07	0.09	0.10	0.05	0.09
China (Mainland)	0.21		0.06	0.12	0.28	0.02	0.93
Germany	0.04	0.08		0.02	0.01	0.01	0.04
Japan	0.07	0.12	0.02		0.01	0.01	0.05
India	0.22	0.11	0.01	0.04		0.04	0.19
United Kingdom	0.05	0.06	0.02	0.01	0.01		0.04
Brazil	0.03	0.06	0.01	0.01	0.02	0.01	

[1]A ratio greater than 1 indicates that rearrangement alone cannot satisfy import demand; however, demand could be met through ramped-up production, etc.
Source: UN Comtrade; McKinsey Global Institute analysis

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Rearrangement ratio calculation: Approach and limitations

Trade rearrangement ratios are calculated at the product level, corresponding to the World Customs Organization's Harmonized System six-digit (HS 6) level. Economy-level rearrangement ratios are calculated as an average weighted by total imports of each product from the given exporter.

Limitations in trade data induce some uncertainty in the value of product-level rearrangement ratios:

- *There is no one "source of truth" on how much countries trade.* Substantial discrepancies sometimes occur between the value of trade that exporting countries report and what importers report. For example, in 2023, China reported exporting $700 million worth of headphones to the United States, while the United States reported importing $3.3 billion worth of headphones from China. The rearrangement ratio varies a lot, depending on which number one chooses. In general, we have used importer-reported figures for the rearrangement ratio numerator, and exporter-reported figures for the denominator. However, this variation in trade flow values suggests the rearrangement ratio for most products can lie within plus or minus 0.2 of the values we report.
- *Re-exports are sometimes misclassified as exports.* Where possible, we have excluded re-exports from our sizing of the global available export market. However, some major re-export hubs feature in the trade data as the exporter, rather than re-exporter, of a product. For example, the Netherlands is often recorded as an exporter of goods that are in fact just transiting through the port at Rotterdam.[1] As a result, the Netherlands features in trade data as the third-largest global exporter of laptops and the fifth-largest smartphone exporter. This phenomenon may inflate the size of the "available" global export market for some products.

In addition, the interpretation of rearrangement ratios is subject to at least two important caveats, which, taken together, will bias some results toward underestimating the rearrangement ratio for a given product:

- *Products within the same HS 6 level may not be substitutable.* Although we use the most granular level of global trade data available, products classified under the same HS 6 classification may nevertheless not be substitutable. This may mask instances where specific products within a broad HS 6 classification have a high rearrangement ratio. Rare earth magnets provide an example. In HS 6 trade data, these magnets are bucketed under "permanent magnets," a category that also includes widely produced and exported non-rare-earth magnets. Rare earth and non-rare-earth magnets are not substitutes in many use cases. The US China-rearrangement ratio for this broader bucket is about 0.25, masking the much higher rearrangement ratio for rare earth magnets. A similar issue of imperfect substitutability within an HS 6 classification exists for other products, which we have not systematically identified.
- *Rearrangement ratios have been calculated using trade values, which are an imperfect proxy for trade volumes.* Trade values are generally more reliably recorded than trade volumes, and therefore our analysis is based on the former. However, using trade values implicitly assumes similar unit prices and consistent price reporting across exporting economies, an assumption that may not generally hold. For example, for a commoditized product like hot-rolled steel band, European exports are on average about 60 percent more expensive than China's. And for something like a cotton T-shirt, the average US import from Europe costs about eight times more than one from China. In these cases, rearranging a certain dollar amount of trade from China to Europe would not deliver the same volume of product.

[1] This form of trade data distortion is often known as the "Rotterdam effect," though this form of trade distortion is not unique to Rotterdam. For an example discussion, see UK Office for National Statistics, "UK Trade in Goods estimates and the 'Rotterdam Effect,'" 2015.

Trade rearrangement approaches: Definitions and limitations

We explore nine simulations of how US imports from China might rearrange. In each simulation, the modeling assumes that the value of US imports from China for each product falls to zero, and that the United States sources each product instead from other exporters.[1] If there is insufficient global supply for the United States to replace China's exports, this will result in a reduction in US imports and a corresponding excess of China's exports. Otherwise, US imports and China's exports will remain at the same value level as in 2023.

In general, these simulations therefore do not substantially affect economies' overall trade balances because rearrangement shifts only who exports to whom, not overall aggregate exports (and imports). For the United States, for example, rearrangement alone leaves the trade deficit unchanged, though the trade deficit with China reduces and shifts to increased deficits with other economies. An exception is the US ramp-up approach—the third simulation described below—in which the US decreases overall aggregate imports.

Six of the nine simulations consider cases where US importers are the "first mover," and trade rearranges in pursuit of a particular goal. They are the following:

— *Rearrangement minimization.* In this case, supply coming from China is replaced by supply from the fewest possible exporters, with no regard to tariff or geopolitical position. With this overarching goal, we examine three distinct simulations:

  - A "US-last" approach. Existing US exports are redirected toward the domestic market only as a last resort, if all other foreign sources of supply have been exhausted (simulation 1).
  - A "US-first" approach. Existing US exports are redirected toward the domestic market before sourcing from overseas (simulation 2).
  - A "US ramp-up" approach. Existing US exports are redirected domestically to the same extent (in dollar terms) as in the "US-first" approach, but production ramps up to maintain US exports of these products at 2023 levels. Note that in this scenario, the overall US trade deficit shrinks by about $200 billion (simulation 3).

  These rearrangement-minimization simulations explore how different US supply responses affect the resulting global rearrangement patterns. Although they may incur a higher tariff bill, these approaches may still make economic sense. Identifying each supplier does require incremental effort to search, specify, and contract, so the effort and cost associated with finding new sources of supply might be lowest in this approach, since it would require fewer countries to replace exports from China.

— *Geopolitical distance minimization.* In this simulation, US imports rearrange toward exporting countries starting with those that are most geopolitically proximate to the United States—starting with domestic supply—and then moving as needed to those that are further away (simulation 4).[2]

— *Tariff minimization.* US imports shift toward exporting countries with the lowest tariffs, starting with the exported supply from the United States (which faces no tariffs) and then moving along the tariff cost curve until China's supply has been replaced. We consider two simulations here and find broadly similar results, indicating that the specifics of the tariffs may not dictate the broad patterns of potential rearrangement. The two tariff simulations we test are the following:

  - US tariffs as announced on April 2, 2025 (simulation 5).[3]
  - As above, except all US imports from the EU face a 50 percent tariff. This explores the role Europe plays in trade rearrangement if it faces higher tariffs than other economies, with the tariff figure reflecting President Donald Trump's statements in late May (simulation 6).

We also consider three cases in which China's exporters are the "first movers," typically analogous to the US simulations. These are the following:

— *Rearrangement minimization.* China rearranges its exports to the fewest possible countries. As with the US equivalent, we consider two cases, as follows:

  - A "China-last" approach. China redirects its exports toward serving domestic demand only if global demand for the product has been saturated (simulation 7).
  - A "China-first" approach. If China imports the same product that it also exports to the United States, it will first redirect this export to serving domestic demand rather than toward a foreign trade partner (simulation 8).

— *Geopolitical distance minimization.* China rearranges its exports to the partners geopolitically closest to it (simulation 9).

[1] The modeling considers the baseline of global trade in 2023, the most recent year for which global data are available. Analyses are conducted at the most detailed available level, the HS 6 trade product classification. Under this classification, just under 6,000 distinct products make up the universe of global goods trade.

[2] Countries' geopolitical positions are approximated using UN General Assembly voting records between 2005 and 2022. This is a backward-looking measure that may not capture recent or ongoing shifts in countries' positions. For a discussion of this approach to measuring geopolitical distance and its limitations, see McKinsey Global Institute, *Geopolitics and the geometry of global trade*, January 2024.

[3] These settings include country-specific tariffs as announced on April 2, 2025, and all prior tariff settings. They also include the April 11, 2025, clarification on exemptions. Note that results of the simulation depend only on the ordering of economies by tariff level for an individual product, and not on the specific tariff levels.

Approach limitations and assumptions. This approach makes several assumptions and has several important limitations. Some of these overlap with those discussed in the context of the rearrangement ratio in the sidebar "Rearrangement ratio calculation: Approach and limitations." Those include the assumption that products within the same HS 6 classification are substitutable and are supplied at similar unit prices by different exporters. In addition, our modeling here does not consider a range of important effects:

— *New sources of production.* Domestic US production or other economies' exports may scale to meet US demand. This would be an increasingly relevant mechanism over the course of months or years (depending on the product), which we have generally not simulated. As an example, when the United States hiked tariffs on washing machines from China in 2016, US imports fell by $650 million, but within a year, an $800 million increase in imports from ramped-up production in Thailand and Vietnam more than compensated.[4]

— *General equilibrium effects.* Shifts in sourcing may lead to price changes, affecting the dynamics of demand and supply within and across sectors and ultimately affecting the allocation of resources and growth. Moreover, especially in the context of tariffs, there may be changes in exchange rates and terms of trade. These important considerations have not been modeled.

— *Practicality.* Most trade relations are the result of long-standing relationships and contracts between exporters and importers. It therefore may not be practical for exports from one economy to rearrange simply to another. To take one example, our modeling assumes that intra-European exports can be redirected toward the United States. In the real world, the extent of this redirection may be limited by additional customs and logistics requirements, supplier and importer "discovery," and so on.

— *Policy responses.* In the face of current trade and broader geopolitical tensions, countries may shift their trade and industrial policies as well as their geopolitical stances. For example, a range of countries might institute new tariffs, or some economies might group together and form new trading blocs. Our simulations do not consider such potential shifts. Changes might also extend to more explicit geopolitical repositioning in a way that shifts future geopolitical distance between countries from what it has been in the past.

[4] This case is discussed in Aaron Flaaen, Ali Hortaçsu, and Felix Tintelnot, "The production relocation and price effects of US trade policy: The case of washing machines," *American Economic Review*, volume 110, number 7, July 2020. They find that there was a corresponding increase in imports of intermediate parts to these economies, suggesting that true new production—rather than "trans-shipment" of goods from China—accounted for the increased production.

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Endnotes

1 In this report, "China" refers to Mainland China unless otherwise specified.

2 For a quantification of product complexity used in the research literature, see the Product Complexity Index, maintained by the Growth Lab at Harvard University.

3 There is no specific threshold at which rearrangement becomes more difficult. Consider truck tires, which had a rearrangement ratio of 0.1 in 2018, before the United States imposed tariffs on China that rose to 25 percent. This led US imports from China to drop by more than 60 percent, but total US truck tire imports remained stable as exports from other economies rearranged toward the United States.

4 These are examples of products for which the granularity of global trade data is insufficient to directly calculate the rearrangement ratios. Rare earth magnets, for instance, are part of a larger category of permanent magnets that have a lower rearrangement ratio. See sidebar "Rearrangement ratio calculation: Approach and limitations" for more detail on the limitations of the rearrangement ratio.

5 See, for example, Glenn Zorpette, "Advanced magnet manufacturing begins in the United States," *IEEE Spectrum*, February 9, 2025.

6 This accords with the UN Broad Economic Categories (5th revision) classification of goods.

7 According to research by the UN Trade and Development (UNCTAD), transporting a product 10,000 kilometers by sea (roughly the distance between Shanghai and Los Angeles) adds, on average, about 5 percent to the cost of a product. Therefore, transportation costs alone are unlikely to substantially shift the tariff cost curve, though their impact will vary by product, mode of transport, and route. See Onno Hoffmeister et al., "Developing a global transport costs dataset for international trade," UNCTAD Research Papers, Number 85, June 2022.

8 Within advanced manufacturing, some sectors like pharmaceuticals, and semiconductors and certain electronics, are excluded from the tariffs announced on April 2, 2025, because Section 232 investigations are underway, with potential for future tariffs or other measures. Energy resources are largely excluded from the announced tariffs.

9 European Union economies plus Norway, Switzerland, and the United Kingdom.

10 Economies in the Other Europe and Central Asia grouping that experience increased exports to the United States include Eastern and Southeastern European countries that are not European Union members (such as Albania and Serbia), Türkiye, and some Central Asian economies.

11 This analysis considers "domestic exports" of goods—those goods that are produced or manufactured domestically—and excludes re-exports. The headline export figure for the United States, about $2 trillion in 2023 and 2024, includes more than $300 million of re-exports.

12 For the purposes of modeling how trade might rearrange, our simulations treat products with the same classification, known as HS 6 codes, as interchangeable. They also assume that trade values broadly reflect trade volumes. We recognize that these assumptions do not always hold. For a fuller discussion of the modeling approach and limitations, see sidebar "Trade rearrangement approaches: Definitions and limitations."

13 For example, in 2024, the US tariff on EV imports from China rose to 100 percent. US imports of solar panels from China have been subject to a series of different tariffs, starting with anti-dumping duties of about 25 percent in 2012. Many steel mill products were subject to both Section 301 tariffs on imports from China in 2018 and the broader Section 232 tariffs on iron and steel imports that were enacted during the first Trump administration.

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